CHIRANJEEV KUMAR

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GitHub | LinkedIn | Portfolio

EDUCATION

Gulzar Group of Institutions — B.Tech in CSE (Specialization: AI/ML), Affiliated to IKGPTU, Jalandhar
Expected Graduation: June 2028
Current CGPA: 7.48 / 10
Relevant Coursework: Data Structures & Algorithms, OOPs, DBMS, Operating Systems, Computer Networks (CN)

TECHNICAL SKILLS

Languages: C++, JavaScript (ES6+), Python, HTML5, CSS3, SQL
Frameworks & Libraries: React.js, Node.js, Express.js, Tailwind CSS, Standard Template Library (STL)
Databases & Developer Tools: MongoDB, Git, GitHub

TECHNICAL PROJECTS

Personal Productivity Progressive Web App (PWA) — Live Demo GitHub
Full-Stack Developer

- Engineered an offline-first productivity workspace that achieved a 40% reduction in perceived task-tracking latency compared to standard blocking network architectures by implementing Optimistic UI updates and background asset compilation
- Implemented an asynchronous outbox synchronization queue using custom Service Workers and local storage snapshots, guaranteeing 100% data preservation and automatic background replay upon network reconnection
- Designed a component-based React architecture with localized state re-validation hooks, enabling modular, reusable UI elements across core task execution, fluid goal tracking, and live-computed calendar modules
- Built secure, stateless RESTful APIs utilizing Node.js, Express.js, and JWT authentication guards, integrated with a cloud-hosted MongoDB cluster for cross-session persistent storage
- **Technologies used:** React.js, Node.js, Express.js, MongoDB, Service Workers, Workbox, JWT, Git

WebAssembly Huffman File Compressor — Live Demo GitHub
Full-Stack Systems Developer

- Built a browser-based, privacy-first file compression tool by compiling a custom C++17 Huffman Encoding engine to WebAssembly via Emscripten, achieving near-native execution speed for client-side compression
- Engineered a cross-language memory bridge using Emscripten's ccall and the MEMFS virtual filesystem to safely transfer raw binary data between JavaScript and the native C++ engine without memory leaks
- Designed a responsive frontend using the FileReader and Blob APIs to process files entirely client-side — no uploads, no server processing, ensuring user data never leaves the device
- Automated cross-compilation and deployment via a GitHub Actions CI/CD pipeline, building optimized .wasm/.js binaries and continuously deploying to Vercel
- Achieved a 49% compression ratio on text data (114 KB → 57.86 KB) using a min-heap-based Huffman tree and bitset-driven bit-packing, validated via byte-for-byte lossless decompression
- **Technologies used:** C++17, WebAssembly, Emscripten, JavaScript (ES6), HTML5/CSS3, GitHub Actions, Vercel, Git

EXPERIENCE & TRAINING

Software Engineer Job Simulation — Skyscanner *(Forage)* Certificate June 2026 – July 2026
Software Engineer | *Ludhiana, Punjab*

- Reconstructed the application architecture to ensure clean separation of components and robust state-handling in React
- Developed a date picker component utilizing Skyscanner's custom Backpack library
- Wrote and executed automated tests using Jest to verify correct application rendering and behavior
- Learned how to debug and optimize build configurations, including package compilation and asset packaging, on native developer environments

ACHIEVEMENTS & CERTIFICATIONS

- **Hackathon Distinction:** Competed against 400+ regional engineering participants to secure a Top 30 Finalist Placement by building custom software logic layers under a 36-hour timeline constraint [Certificate]
- **Cloud Consultant Certification** — PMKVY (NSQF Level 6), IT-ITeS Sector Skill Council, NASSCOM-recognized; 600 hrs, Grade A [Certificate]
- AWS IAM & EC2 hands-on lab session — Cloud Clubs, Gulzar Group of Institutes [Certificate]